An amplifier without a signal: denervation supersensitivity as a mechanism for hard flaccid syndrome

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[1] *HFS Patient* • [2] *Large language model, Anthropic PBC*

Sixth in a series. Companion papers: *The sacral splanchnic bypass*; *Initiation without maintenance*; *The intraganglionic loop*; *A generator that outlives its cause*; *Autonomy without a signal*.

Abstract

Background. If sympathetic blockade genuinely fails to abolish the hard flaccid state, the effector tissue must have become autonomously hypercontractile. One candidate route is denervation supersensitivity: partial loss of autonomic innervation rendering cavernosal smooth muscle abnormally responsive, so that normal or even reduced catecholamine exposure produces exaggerated contraction.

Hypothesis. We propose that the initiating injury in hard flaccid syndrome (HFS) partially denervates the cavernosal tissue, and that the resulting adaptive supersensitivity converts ordinary physiological catecholamine exposure — including the circulating catecholamines that no regional nerve block removes — into a symptomatic elevation of resting tone.

A strength the conjecture has not claimed. Adaptive supersensitivity in smooth muscle is not primarily a receptor phenomenon. It is nonspecific, extends to agonists unrelated to the lost transmitter, occurs without change in receptor density or affinity, and correlates with partial depolarisation of the smooth muscle cell caused by reduced electrogenic sodium–potassium pumping. *A partially depolarised smooth muscle cell has elevated resting tone in its own right.* The mechanism therefore does not merely amplify a signal; it raises baseline tone directly, which is what the hypothesis actually needs and what pure receptor upregulation would not deliver.

The strongest argument against it. Radical prostatectomy is the largest human model of cavernosal autonomic denervation in existence, involving very large numbers of well-characterised patients. Its recognised sequelae are erectile dysfunction, penile shortening and corporal fibrosis — not a hard, hypercontractile flaccid penis. The cavernous nerve injury literature reports smooth muscle apoptosis, phenotypic modulation from contractile toward synthetic, and collagen deposition: changes that reduce contractile capacity rather than increase it. We regard this natural experiment as the most serious obstacle the hypothesis faces, and set out the three ways it might be answered.

Tests. Dilute-agonist supersensitivity testing is an established human diagnostic method with published quantitative norms, sensitivity and specificity, used routinely in autonomic neuro-ophthalmology. We adapt it: graded intracavernosal α-agonist dose–response against age-matched controls as the direct test, and cutaneous iontophoretic agonist challenge with laser Doppler flowmetry as a less invasive surrogate.

Conclusion. The hypothesis rests on a classical and well-replicated physiological phenomenon whose smooth-muscle mechanism is better suited to this disorder than has been recognised. It is opposed by the outcome of the one large human denervation experiment we have already run.

Keywords: hard flaccid syndrome; denervation supersensitivity; adaptive supersensitivity; corpus cavernosum; sodium–potassium pump; circulating catecholamines; effector autonomy.

Reader advisory

This is hypothesis generation, not a report of new data. No component of the mechanism proposed here has been demonstrated in a person with hard flaccid syndrome. The principal test proposed involves administering a contractile agent into the penis, which carries real risk and would require careful ethical justification; it is described as a research procedure and is emphatically not a treatment. Nothing here should influence clinical management.

1. Introduction

Four papers in this series place the lesion in the nervous system [5, 6, 7, 8]; a fifth places it in the contractile machinery of the smooth muscle cell [9]. All six are answers to the same problem. If a technically confirmed sympathetic block genuinely fails to abolish the hard flaccid state — and the reports that it does are unverified, undenominated and undocumented **[E]** [4, 5] — then the contraction is not being sustained by ongoing neural traffic, and something in the effector tissue must have changed.

This paper examines a different route to that conclusion, and one with an unusual property: it explains the block failure *without requiring the neural signal to be abnormal at all*. Under denervation supersensitivity, the sympathetic outflow to the penis may be entirely normal. What has changed is the tissue's response to it — and, critically, its response to the circulating catecholamines that no regional nerve block removes.

The idea is old. Cannon and Rosenblueth formalised it as a law of denervation: when a unit is destroyed in a chain of efferent neurons, an increased irritability to chemical agents develops in the isolated structure or structures **[C]** [VERIFY] [10]. It has been characterised in detail in smooth muscle over six decades, and — a point that matters for feasibility — it is tested clinically in humans every day, though not in this organ.

1.1 What this paper adds to the conjecture as stated

Two things. First, the conjecture as usually stated treats supersensitivity as receptor upregulation producing an exaggerated response to agonist [4]. That is the skeletal muscle picture. In smooth muscle the mechanism is different and, for present purposes, better: it is electrophysiological, involves partial depolarisation of the cell, and therefore raises resting tone directly rather than only amplifying responses (Section 3.2). Second, the conjecture identifies its own weakness as the question of whether a somatic sensory nerve injury can denervate autonomic fibres. We think that objection is answerable (Section 3.4) — and that a different and much harder objection has been missed (Section 5).

2. Methods

This is a hypothesis and theory article. No new human or animal data were generated; no patients were involved; ethical approval was not required.

Literature was identified by targeted searching of MEDLINE/PubMed and the open web between May and August 2026, using combinations of the terms *denervation supersensitivity*, *adaptive supersensitivity smooth muscle*, *Trendelenburg supersensitivity sympathomimetic*, *dilute pilocarpine tonic pupil*, *cavernous nerve injury smooth muscle*, and *hard flaccid syndrome*, with backward citation chasing. The search was purposive and non-systematic; no protocol was registered and no formal risk-of-bias appraisal was applied. We searched specifically for evidence of adrenergic supersensitivity in penile tissue after denervation, and Section 6 reports that we found none.

Claims are graded as in Table 1. Claims not verified against a retrievable source are marked [VERIFY]; quantities that do not exist in the literature are marked [DATA NEEDED].

Table 1. Evidence grading scheme. The grade refers to the strength of evidence for the *stated proposition*, not to its relevance to hard flaccid syndrome. No proposition here carries grade A evidence *in HFS*, because none exists.

Grade	Definition
[A]	Replicated human evidence: consistent findings across independent human studies.
[B]	Single human study, human tissue study, or human data with substantial methodological limitation.
[C]	Experimental animal or in vitro evidence, with species and preparation stated.
[D]	Mechanistic inference: deduced from A–C evidence but not itself directly observed.
[E]	Unverified anecdotal or community-sourced report; not peer reviewed, denominators unknown.

3. Background

3.1 Two components, and which lesions produce them

Trendelenburg distinguished two forms of supersensitivity to sympathomimetic amines. One is produced by cocaine: rapid in onset and highly specific, affecting only amines subject to neuronal uptake. The other is produced by decentralisation — preganglionic denervation — or by reserpine: slow in onset and non-specific. Postganglionic denervation, which destroys the nerve terminals themselves, produces *both* components together **[C]** [VERIFY] [11, 14].

The distinction matters for HFS because it determines what kind of lesion is required. A lesion that destroys terminal noradrenergic axons within the penis would produce both the prejunctional component — loss of neuronal reuptake, raising the concentration of transmitter reaching the receptor — and the postjunctional component. A lesion sparing the terminals would produce only the second **[D]**.

3.2 In smooth muscle, the mechanism is electrophysiological, not receptor-based

This is the most important thing in this paper and it is not what the conjecture assumes.

Adaptation of smooth muscle to denervation differs from that of skeletal muscle. The supersensitivity extends to agonists unrelated to the lost neurotransmitter; it develops over days to weeks; it is elicited by several procedures besides denervation that interrupt excitatory transmission; and — decisively — it occurs *without changes in receptor density or affinity*, correlating instead with partial depolarisation of the smooth muscle cells. The depolarisation is attributed to reduced electrogenic pumping and reduced density of the sodium–potassium pump, and the phenomenon can be mimicked by procedures that acutely depolarise the cells **[C]** [VERIFY] [13, 12].

Two consequences follow, and the second is the one that makes this hypothesis viable **[D]**:

1. **The supersensitivity is nonspecific.** It is not confined to α-adrenoceptor agonists. This directly predicts what the conjecture claims it explains: that α-blockade would only partially reverse the state, because blocking one receptor class does nothing about sensitivity to the others.
2. **Partial depolarisation raises resting tone by itself.** A smooth muscle cell held closer to threshold has greater basal calcium entry through voltage-gated channels and therefore greater basal tone, independently of any agonist. This answers an objection that would otherwise be fatal: pure receptor upregulation makes a tissue *more responsive* without making it *more contracted at rest*, and HFS is a disorder of resting tone. The electrophysiological mechanism delivers both.

We emphasise that the second point is our inference from the reported mechanism, not a demonstrated finding in cavernosal tissue, and that the magnitude of depolarisation required to produce a clinically evident change in penile tone has never been estimated [DATA NEEDED].

3.3 Supersensitivity testing is established clinical practice — in another organ

The hypothesis proposes a phenomenon that clinicians already diagnose routinely, using dilute agonists and a quantitative response threshold. In unilateral Adie's tonic pupil, dilute pilocarpine demonstrates cholinergic supersensitivity of the iris sphincter; in a case–control study of 117 subjects, a change in maximal pupil diameter of at least 0.5 mm after 0.0625% pilocarpine gave 100% sensitivity and 82.8% specificity, with an area under the receiver operating characteristic curve of 0.954 — significantly better than the conventionally recommended 0.125% concentration (AUC 0.840, $p = 0.047$) **[B]** [16]. An earlier standardised protocol established quantitative normative values for iris responses to dilute phenylephrine and pilocarpine, including an age correction, and reported supersensitivity in 71% of sympathetically denervated or decentralised (Horner) pupils **[B]** [VERIFY] [17].

Three lessons transfer directly to any test designed for the penis **[D]**. **Concentration matters more than the presence of an effect** — the more dilute agonist performed better, because supersensitivity is a leftward shift and a strong agonist saturates both groups. **Age-adjusted normative data are required**, since sensitivity drifts with age. And **even in a definite denervation state the test is not universally positive**: 71% in Horner pupils, meaning a negative result in an individual patient would not exclude the mechanism.

3.4 Can a penile stretch injury denervate autonomic fibres?

The conjecture identifies this as its own weakness: a stretch injury to the dorsal nerve of the penis — a somatic sensory branch — is not obviously a sympathetic denervating lesion [4].

The objection is answerable. The dorsal nerve of the penis is not purely somatic. Postganglionic sympathetic fibres reach it directly from the sympathetic chain: stimulation of the lumbar sympathetic chain at L4–L5 in the rat elicited evoked discharge on the dorsal nerve of the penis, abolished by cutting the chain distal to the stimulation site, and ganglionic blockade with serial neural sections demonstrated postganglionic sympathetic fibres within the nerve **[C]** [18]. In humans, sympathetic fibres are found in the pelvic, cavernous *and pudendal* nerves **[B]** [19], and communicating branches among the dorsal, perineal and cavernous nerves of the penis have been demonstrated anatomically **[B]** [VERIFY] [20].

More importantly, a violent stretch of the whole organ is not a lesion of one named nerve. It would act on everything running along the shaft, including the terminal branches of the cavernous nerves and the perivascular noradrenergic plexus within the erectile tissue itself. **If it denervates anything autonomic, it denervates it at the terminal — that is, postganglionically — which is precisely the lesion that produces both components of supersensitivity [D]**.

This repairs the conjecture's stated weakness. It does not repair the one it has not stated.

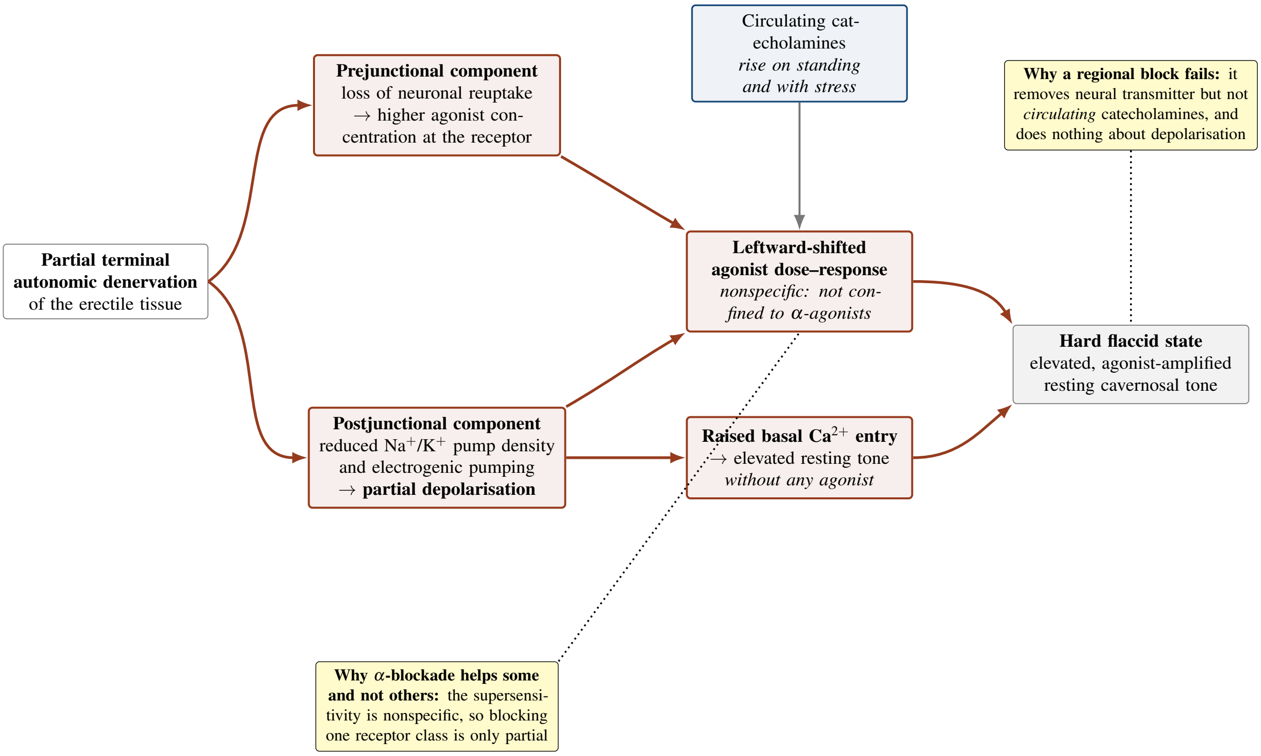

Figure 1. The two components of adaptive supersensitivity and their proposed consequences. A lesion destroying terminal noradrenergic axons produces both components. The prejunctional component (loss of neuronal reuptake) raises the concentration of transmitter — and of circulating catecholamine — reaching the receptor. The postjunctional component in smooth muscle is not receptor upregulation but reduced electrogenic sodium–potassium pumping causing partial depolarisation, which both shifts the agonist dose–response leftward and, independently, raises basal calcium entry and resting tone. The second of these is what the hypothesis requires and what a purely receptor-based account would not supply. Diagram is schematic; the depolarisation mechanism is reported for smooth muscle generally [13] and has not been examined in cavernosal tissue.

4. The hypothesis

Formal statement

H6 (denervation supersensitivity). The initiating injury in hard flaccid syndrome partially denervates the cavernosal tissue at the terminal level. The resulting adaptive supersensitivity — prejunctional loss of reuptake plus postjunctional partial depolarisation — raises resting tone and amplifies the response to catecholamines, so that ordinary physiological exposure produces a symptomatic state.

H6-humoral (the form that explains block failure). The state is sustained substantially by circulating catecholamines acting on a supersensitive, reuptake-deficient tissue. Prediction: no regional sympathetic block abolishes it; systemic α-blockade does better than regional blockade; symptoms track systemic sympathoadrenal activity.

H6-intrinsic. The postjunctional depolarisation raises tone without requiring any agonist. Prediction: elevated tone persists under full adrenoceptor blockade; the tissue is supersensitive to non-adrenergic contractile agonists as well.

H6-null. Cavernosal sensitivity to contractile agonists in HFS is normal. Whatever maintains the state is neural, structural, or not present.

Auxiliary assumptions:

- **A1 (denervating lesion).** A penile stretch injury damages terminal autonomic axons. Anatomically plausible (Section 3.4); never demonstrated [DATA NEEDED].
- **A2 (persistence).** The supersensitive state persists for years. Classical adaptive supersensitivity develops over days to weeks and is expected to regress with reinnervation [13]; persistence over years requires either failed reinnervation or a maintaining process not specified by the hypothesis. **This is a substantial and under-acknowledged problem.**
- **A3 (magnitude).** The degree of supersensitivity is sufficient to convert ordinary catecholamine exposure into symptomatic tone. Classical supersensitivity in smooth muscle produces a leftward shift of modest size; whether that is enough has never been estimated for this tissue [DATA NEEDED].

5. The natural experiment that argues against it

We have already performed the experiment this hypothesis proposes, in very large numbers, and the result does not look like hard flaccid syndrome.

Radical prostatectomy injures the cavernous nerves. It is the largest and best-characterised human model of cavernosal autonomic denervation in existence, and its consequences have been studied intensively for decades. The recognised sequelae are erectile dysfunction, loss of nocturnal erections, penile shortening and corporal fibrosis **[A]** [VERIFY] [23, 22]. **A persistently hard, hypercontractile flaccid penis is not among them.**

The animal literature points the same way. In bilateral cavernous nerve injury models, the reported changes in cavernosal smooth muscle are apoptosis, phenotypic modulation from a contractile toward a synthetic phenotype with reduced α-smooth muscle actin and calponin, hypoxia, and an increased ratio of collagen to smooth muscle **[C]** [VERIFY] [21, 22, 23]. These are changes that *reduce* contractile capacity. If autonomic denervation of the corpora produced clinically significant hypercontractility, this literature is where it would have appeared, and it has not.

5.1 Three ways the objection might be answered

We set them out because the hypothesis should not be dismissed on a single argument, and because each generates a testable difference.

1. **Wrong level.** Prostatectomy injures the cavernous nerve trunk proximally — a lesion that is postganglionic but not *terminal*. HFS would require injury to the intracavernosal terminal plexus. Whether terminal and trunk lesions produce the same adaptive response in this tissue is unknown [DATA NEEDED]. This is the most substantial of the three answers, and it is testable in an animal model.

2. **Wrong completeness.** Prostatectomy denervation is extensive and often bilateral; supersensitivity requires surviving effector tissue and, for the humoral form, a functioning contractile apparatus. Extensive denervation followed by apoptosis and fibrosis may destroy the substrate before supersensitivity can express itself. A partial lesion might sit in a window that a complete one overshoots **[D]**.

3. **Nobody has looked.** Hard flaccid syndrome was first described in the peer-reviewed literature in 2019–2020 [1]; is unfamiliar to a large proportion of sexual medicine practitioners **[B]** [2]; and has no diagnostic criteria. A post-prostatectomy patient reporting a firm flaccid penis would have been recorded as having erectile dysfunction. **This is a real possibility and it is also cheaply checkable**: prospective symptom screening for hard flaccid features in a post-prostatectomy cohort would either produce cases or largely close the question (Section 7).

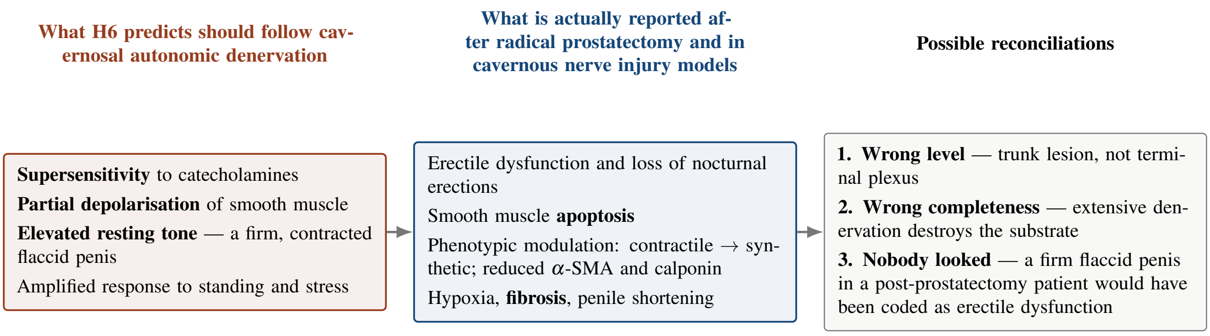

Figure 2. The radical prostatectomy natural experiment. Cavernosal autonomic denervation has been performed in very large numbers of men and studied intensively. The observed phenotype is loss of contractile tissue, not gain of contractile tone. Reconciliation 3 is the only one that is cheaply testable in humans, and it is the study we recommend most strongly: prospective screening of a post-prostatectomy cohort for hard flaccid features. A negative result would substantially weaken this hypothesis; a positive result would be a significant finding in its own right, independent of any theory.

6. What has not been looked for

We identified no study measuring adrenergic sensitivity, dose–response relationships, resting membrane potential, or sodium–potassium pump density in cavernosal tissue after denervation, in any species [DATA NEEDED]. The cavernous nerve injury field has focused almost exclusively on failure of erection and on the structural changes that accompany it. The question this hypothesis asks — does denervated cavernosal smooth muscle become supersensitive? — appears never to have been posed.

That is a genuine gap and, unlike most gaps identified in this series, it is one that could be closed in an existing animal model with established methods and without any new technology. Organ-bath dose–response curves for phenylephrine and for a non-adrenergic contractile agonist, in corpora from cavernous-nerve-injured and sham animals, would answer it. If denervated cavernosal tissue does not become supersensitive, the hypothesis fails at the first step **[D]**.

6.1 Two further pieces of unfavourable evidence

Supersensitivity is tissue-dependent and not universal. In rat mesenteric arteries, chronic interruption of adrenergic transmitter release produced prejunctional supersensitivity but gave no indication of postjunctional supersensitivity **[C]** [VERIFY] [15]. Whether cavernosal smooth muscle behaves like the cat nictitating membrane, in which the phenomenon is robust, or like the rat mesenteric artery, in which it is not, is unknown.

The quantitative case is unmade. For the humoral form of the hypothesis, circulating catecholamines must reach effective concentrations at cavernosal α-adrenoceptors. Resting plasma noradrenaline concentrations are far below those achieved at the neuroeffector junction during nerve activity, and we could not establish from the literature whether the leftward shift produced by adaptive supersensitivity, combined with loss of neuronal uptake, is sufficient to bridge that gap [DATA NEEDED]. This calculation has not been done, and it should be done before the humoral form is taken seriously.

7. Predictions and tests

7.1 The cheap test that addresses the strongest objection

Prediction P1. Prospective screening of a post-radical-prostatectomy cohort using a standardised hard flaccid symptom instrument will identify cases with a persistently firm, non-compressible flaccid penis at a frequency above the background population rate.

This requires no new procedure, no drug and no laboratory. It uses a population that is already followed systematically. It addresses the hypothesis's most serious problem directly, and it is informative in both directions: a positive result would be a substantive clinical finding independent of any theory, and a negative result would substantially weaken H6. **We regard this as the highest-value study proposed in this paper.** Its limitation is that no validated HFS symptom instrument exists, so one would have to be constructed first [DATA NEEDED].

7.2 The direct test: graded intracavernosal agonist dose–response

Prediction P2. Men with HFS show a leftward-shifted dose–response to an intracavernosal α-adrenoceptor agonist compared with age-matched controls.

The design should follow the lessons of the ophthalmic supersensitivity literature (Section 3.3): use *dilute* concentrations rather than clinical doses, because a strong agonist saturates both groups and abolishes the difference; establish age-adjusted normative values, since receptor sensitivity drifts with age; and pre-specify a quantitative response threshold rather than a subjective impression [16, 17].

Feasibility and safety must be stated honestly. Phenylephrine is administered intracavernosally in established clinical practice, so the route and the agent are not novel and dosing precedents exist. But administering a contractile agent to a young man whose complaint is excessive contraction is ethically uncomfortable, requires careful justification, and demands a pre-specified stopping rule and reversal plan. The outcome measure is also missing: no validated objective measure of cavernosal tone exists [5, 9] [DATA NEEDED]. We describe this as a research proposal only.

7.3 A less invasive surrogate

Prediction P3. Iontophoretic or topical application of a dilute α-agonist to penile skin produces greater vasoconstriction, measured by laser Doppler flowmetry or perfusion index, in men with HFS than in matched controls.

This tests cutaneous rather than cavernosal vascular smooth muscle, which is a real limitation — but if the denervating lesion is a stretch injury to the whole organ, cutaneous vessels lie within the injured territory. It is non-invasive, blindable, contralateral- or adjacent-site-controllable, and directly analogous to the established ophthalmic protocol. We would do this before P2.

7.4 Predictions that discriminate H6 from its siblings

Prediction P4 (nonspecificity). Because supersensitivity in smooth muscle extends to unrelated agonists, HFS tissue should be supersensitive to non-adrenergic contractile agonists as well. **No other conjecture in this series predicts this**, and it is the sharpest available discriminator between denervation supersensitivity and, for example, calcium sensitization [9], which would also produce a leftward shift but through a different final common path.

Prediction P5 (systemic beats regional). Systemic α-adrenoceptor blockade should outperform regional sympathetic blockade, because the former reaches circulating catecholamine effects and the latter does not. This prediction is shared with one resolution of the central-maintenance hypothesis [6] and does not discriminate between them.

Table 2. Predictions of H6 compared with the principal alternatives.

Observation	H6 (supersensitivity)	Calcium sensitization	Neural drive (any route)	Discriminating value
Supersensitivity to *non-adrenergic* contractile agonists	**Predicted**	Not predicted	Not predicted	**High**
Leftward-shifted α-agonist dose–response	**Predicted**	Predicted	Not predicted	Moderate
Confirmed sympathetic block fails to reduce tone	Predicted	Predicted	Not predicted	Moderate
Cutaneous adrenergic supersensitivity in penile skin	**Predicted**	Not predicted	Not predicted	Moderate–high
Hard flaccid features in post-prostatectomy cohorts	**Predicted**	Not predicted	Not predicted	**High, and currently unobserved**
Reduced smooth muscle content, apoptosis, fibrosis after denervation	**Not predicted**	Not predicted	Not predicted	**High, against H6**
Symptoms track systemic sympathoadrenal activity	Predicted	Not predicted	Predicted	Low
Partial and variable response to α-blockade	Predicted (nonspecificity)	Predicted	Not predicted	Low

8. Alternative explanations

1. **Technical failure of the blocks.** The observation motivating the whole effector-autonomy class is grade E and undocumented [4, 5]. If the blocks were inadequate, there is no autonomy to explain.

2. **Calcium sensitization.** RhoA/Rho-kinase upregulation would produce a leftward-shifted contractile response by a different route [9]. It is distinguished from H6 by nonspecificity across agonist classes and by whether the shift persists under full adrenoceptor blockade.

3. **The opposite lesion.** The cavernous nerve injury literature suggests that autonomic denervation of the corpora produces loss of contractile smooth muscle. If HFS follows a denervating injury at all, the expected consequence is a soft, fibrotic, poorly erectile penis — which is closer to the late-stage picture some patients describe than to the early one [4]. **It is possible that H6 has the direction of effect exactly backwards.**

4. **Reinnervation, not denervation.** Adaptive supersensitivity regresses with reinnervation [13]. A disorder lasting years in young men with intact vasculature would require reinnervation to fail persistently, which is not the usual outcome of a stretch injury.

5. **The premise may be false.** As in every paper in this series: cavernosal tone has never been objectively shown to be elevated in HFS, and sympathetic outflow to the penis has never been measured in an affected person.

9. Implications

No clinical implications are drawn. Three research implications follow.

First, screen a post-prostatectomy cohort. The hypothesis's hardest problem is also its cheapest test, and the answer would be clinically interesting whichever way it came out.

Second, do the organ-bath experiment. Whether denervated cavernosal tissue becomes supersensitive has apparently never been asked, and it is answerable in an existing animal model with established methods. Nothing else in this paper is worth doing until it is.

Third, use dilute agonists. If any human supersensitivity test is attempted, the ophthalmic literature is unambiguous that concentration selection determines whether the test works at all. A study using clinical-strength phenylephrine would probably return a false negative.

10. Limitations

1. **No evidence in HFS.** Adrenergic sensitivity has never been measured in a patient with this condition.

2. **No evidence in the tissue, in any species.** We found no study of adrenergic supersensitivity in cavernosal tissue after denervation. The first step of the hypothesis is untested.

3. **The natural experiment points the wrong way.** Cavernosal autonomic denervation in humans is associated with loss of contractile tissue, not gain of tone (Section 5).

4. **Persistence is unexplained.** Adaptive supersensitivity develops over days to weeks and regresses with reinnervation; a disorder lasting years requires a maintaining process the hypothesis does not specify.
5. **The quantitative case for the humoral form has not been made**, and we were unable to establish whether circulating catecholamines could reach effective concentrations even in a supersensitive, reuptake-deficient tissue.
6. **Tissue-dependence.** At least one vascular preparation showed no postjunctional supersensitivity after chronic transmitter interruption [15].
7. **The key mechanistic claim is inferred.** That partial depolarisation raises resting tone in cavernosal smooth muscle is our inference from the general smooth muscle literature, not a finding in this tissue.
8. **The direct human test is ethically awkward and lacks an endpoint**, requiring administration of a contractile agent to men complaining of excessive contraction, with no validated measure of the outcome.
9. **Non-systematic search**, with a substantial proportion of citations verified only against secondary sources; all are marked [VERIFY]. Several concern classical mid-twentieth-century pharmacology that we could not retrieve in full.
10. **Confirmation risk.** This paper was written to develop a specific conjecture. We have tried to counter it by identifying an objection the conjecture does not raise against itself, by reporting that the central mechanistic question has never been asked in this tissue, and by naming the possibility that the direction of effect is reversed. The reader should still discount accordingly.

11. Refutation criteria

- **Substantially refuted.** Organ-bath studies show that cavernosal tissue from denervated animals is not supersensitive to contractile agonists relative to sham controls.
- **Substantially refuted.** Graded dilute intracavernosal α-agonist challenge shows no leftward shift in men with HFS relative to age-matched controls.
- **Substantially weakened.** Prospective screening of a post-prostatectomy cohort finds no excess of hard flaccid features.
- **Substantially weakened.** Supersensitivity, if demonstrated, is confined to α-adrenoceptor agonists — indicating receptor-specific upregulation rather than the nonspecific adaptive phenomenon, and removing the explanation for variable α-blocker response.
- **Weakened.** A technically confirmed sympathetic block reliably abolishes objectively measured resting tone, removing the effector-autonomy problem the hypothesis exists to solve.
- **Rendered moot.** Direct measurement shows resting cavernosal tone is not objectively elevated in symptomatic patients.

12. Conclusion

Denervation supersensitivity is the most classical mechanism invoked anywhere in this series — described as a law of denervation three quarters of a century ago, characterised in smooth muscle over six decades, and tested at the bedside every day in a different organ. Applied to hard flaccid syndrome it has a distinctive virtue: it explains why a sympathetic block might fail *without requiring the neural signal to be abnormal at all*. And the smooth-muscle form of the mechanism is better suited to the disorder than the conjecture has claimed. It is not receptor upregulation. It is partial depolarisation from reduced sodium–potassium pumping, which raises resting tone directly rather than merely amplifying a response — and resting tone is what this condition is about.

Against that stand three things. The mechanistic question has apparently never been asked in cavernosal tissue in any species, so the hypothesis's first step is not merely unproven but unexamined. Adaptive supersensitivity develops in weeks and regresses with reinnervation, while the condition lasts years. And most seriously, we have already run the human denervation experiment: radical prostatectomy injures the cavernous nerves in very large numbers of men, and produces apoptosis, phenotypic modulation, fibrosis and erectile failure — loss of contractile tissue, not gain of contractile tone. A hard flaccid penis is not a recognised outcome.

That last objection may be answerable. It may be that a terminal lesion differs from a trunk lesion, that a partial lesion sits in a window a complete one overshoots, or simply that nobody has looked in a population where the complaint would have been recorded as erectile dysfunction. The third possibility is testable next year, in a cohort that is already being followed, with a questionnaire that does not yet exist but would take a season to build. It is the cheapest study proposed in this series and it interrogates the hypothesis at its weakest point. If post-prostatectomy men are not developing hard flaccid syndrome, this conjecture is in serious trouble. If some of them are, that finding matters regardless of whether anyone believes the theory that prompted the search.

Declarations

Authorship and the use of artificial intelligence. The second listed author is a large language model. It is listed at the corresponding author's request and in the interest of transparency, but this contravenes the recommendations of the International Committee of Medical Journal Editors, which hold that artificial intelligence tools cannot meet authorship criteria because they cannot take responsibility for the accuracy and integrity of the work. **Before submission to any peer-reviewed venue this attribution should be converted into an AI-use disclosure, with the human author accepting sole responsibility for the content.** Every citation, and particularly those marked [VERIFY], should be checked against the primary source first; the classical pharmacology references in particular were traced through secondary reference lists and several could not be retrieved in full.

Ethics approval. Not applicable. No human participants, animals or identifiable data were involved.

Funding. None.

Competing interests. No competing financial interests. The first author maintains a public website on hard flaccid syndrome and authored the source article [4]; this is a non-financial interest in the hypothesis being taken seriously, and readers should weigh the argument accordingly.

Data availability. No datasets were generated or analysed.

Clinical disclaimer. This article is hypothesis generation. It does not constitute medical advice and does not establish a clinician–patient relationship. Nothing in it should be used to select, request or avoid any treatment. Intracavernosal injection carries real risk, must never be self-administered, and the agonist challenge described here is a research procedure that has never been performed in this population. Seek emergency care for an erection lasting more than four hours.

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